

INVERGORDON HARBOUR, United Kingdom

WMO: 030580

Lat: 57.683N

Lon: 4.167W

Elev: 10

StdP: 14.69

Time Zone: 0.00 (EUW)

Period: 90-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	27.4	29.7	23.2	17.5	32.1	25.0	19.1	33.8	33.4	44.9	30.0	46.3	3.6	280	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.1	69.8	61.4	67.1	60.0	65.0	58.8	62.9	67.8	61.1	65.7	59.6	64.0	10.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.6	78.9	65.2	58.8	74.0	63.9	57.3	70.1	62.5	28.4	68.3	27.2	65.9	26.2	64.2	68.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.7	25.7	22.4	DB	23.6	74.8	4.9	4.0	20.0	77.6	17.2	80.0	14.4	82.2	10.8	85.1	(4)
(5)			WB	23.0	65.5	4.7	1.7	19.6	66.8	16.8	67.8	14.2	68.7	10.7	70.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	39.7	40.7	43.0	45.3	50.4	54.4	58.1	58.6	54.5	48.9	43.6	39.5	(6)
(7)		DBStd	8.13	4.48	5.21	4.48	4.37	4.64	3.76	3.44	3.77	3.88	4.47	5.23	5.45	(7)
(8)		HDD50	1613	319	262	219	151	50	7	0	0	7	73	198	327	(8)
(9)		HDD65	6169	784	679	681	592	453	318	214	200	315	500	642	791	(9)
(10)		CDD50	921	0	2	3	9	62	139	252	268	142	37	6	1	(10)
(11)		CDD65	5	0	0	0	0	0	0	2	3	0	0	0	0	(11)
(12)		CDH74	11	0	0	0	0	0	0	5	5	1	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	9.5	10.4	12.3	11.7	10.0	8.7	8.8	8.2	7.8	8.3	9.4	8.9	9.6	(14)
(15)	Precipitation	PrecAvg	34.8	3.5	3.2	3.0	2.4	2.3	2.6	2.0	2.3	3.1	3.6	3.4	3.2	(15)
(16)		PrecMax	45.6	6.9	6.5	6.5	4.3	4.7	5.2	3.7	4.8	9.7	6.0	4.9	5.7	(16)
(17)		PrecMin	26.3	0.5	1.0	1.3	1.3	1.1	1.0	0.9	0.8	1.3	0.9	0.7	0.4	(17)
(18)		PrecStd	4.9	1.9	1.7	1.5	0.8	1.0	1.3	0.8	1.1	1.9	1.3	1.1	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	54.8	55.0	58.6	67.2	69.7	74.1	73.0	70.1	63.3	57.9	54.0	(19)
(20)			MCWB	48.7	49.7	49.2	51.3	57.3	58.5	63.7	63.0	61.9	56.7	53.3	50.6	(20)
(21)		2%	DB	50.5	53.2	53.1	56.1	63.6	65.6	68.7	70.3	65.2	59.5	55.6	51.9	(21)
(22)			MCWB	46.5	48.9	48.0	49.6	55.2	57.4	60.9	62.5	59.0	54.5	51.4	48.1	(22)
(23)		5%	DB	48.7	51.5	51.6	54.1	60.4	63.1	66.2	67.5	62.5	57.3	53.1	50.3	(23)
(24)			MCWB	44.9	47.6	47.1	48.0	53.4	55.5	60.1	60.9	57.1	52.7	49.3	46.8	(24)
(25)		10%	DB	46.6	49.4	50.0	52.3	57.8	60.9	64.2	65.0	60.6	55.5	51.2	47.7	(25)
(26)			MCWB	43.3	45.7	46.0	47.3	51.6	53.9	59.1	59.3	55.5	51.1	47.9	44.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.5	51.1	50.3	52.5	58.2	60.5	64.4	66.1	62.2	57.2	53.7	51.0	(27)
(28)			MCDWB	52.7	54.0	54.0	57.5	66.1	66.4	71.2	70.4	67.4	62.6	57.2	53.4	(28)
(29)		2%	WB	46.9	49.4	48.8	50.3	56.2	58.5	62.4	63.6	59.8	55.3	51.7	48.3	(29)
(30)			MCDWB	50.0	52.7	52.5	54.8	62.6	64.3	66.8	68.0	64.1	59.1	55.3	51.2	(30)
(31)		5%	WB	45.1	47.7	47.6	49.0	54.0	56.7	60.9	61.7	58.0	53.5	49.7	47.0	(31)
(32)			MCDWB	48.3	51.2	51.2	53.0	59.3	62.0	65.3	66.6	61.6	56.8	52.5	50.0	(32)
(33)		10%	WB	43.4	45.6	46.1	47.8	52.2	55.1	59.5	60.0	56.5	51.8	48.2	44.8	(33)
(34)			MCDWB	46.5	49.4	49.7	51.5	56.9	59.5	63.5	64.3	59.5	54.8	51.0	47.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.9	7.1	8.2	9.1	10.3	9.7	9.5	10.1	9.6	8.3	7.1	6.8	(35)
(36)			MCDBR	8.9	7.5	9.8	11.9	14.6	13.6	12.6	13.6	12.7	10.9	8.8	9.5	(36)
(37)			MCWBR	7.7	6.2	7.1	7.7	8.4	7.7	7.4	7.6	8.5	7.9	7.3	8.1	(37)
(38)		5% WB	MCDWB	8.8	7.8	9.6	10.5	14.0	12.8	11.8	12.9	11.3	10.2	8.5	9.0	(38)
(39)			MCWBR	7.8	6.8	7.6	7.6	8.7	7.9	7.5	8.1	8.3	8.2	7.5	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.317	0.340	0.363	0.383	0.384	0.379	0.382	0.368	0.355	0.340	0.318	0.305	(40)
(41)		taud		2.202	2.283	2.288	2.277	2.325	2.372	2.377	2.422	2.456	2.404	2.257	2.144	(41)
(42)		Ebn at Noon		164	219	247	262	270	272	269	265	251	218	163	128	(42)
(43)		Edh at Noon		16	23	30	36	36	35	35	31	26	21	16	13	(43)
(44)	All-Sky Solar Radiation	RadAvg		122	316	641	1076	1451	1420	1348	1072	744	390	163	84	(44)
(45)		RadStd		17	37	59	87	100	132	123	68	60	39	15	10	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page